

# Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects\*

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## Abstract

To estimate the dynamic effects of an absorbing treatment, researchers often use two-way fixed effects regressions that include leads and lags of the treatment. We show that in settings with variation in treatment timing across units, the coefficient on a given lead or lag can be contaminated by effects from other periods, and apparent pretrends can arise solely from treatment effects heterogeneity. We propose an alternative estimator that is free of contamination, and illustrate the relative shortcomings of two-way fixed effects regressions with leads and lags through an empirical application.

*Keywords:* difference-in-differences, two-way fixed effects, pretrend test

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following differences. First, in the setting where there is no never-treated group, our method uses the last cohort to be treated as a control group, whereas Callaway and Sant’Anna (2020a) use the set of not-yet-treated cohorts. Our method and theirs thus rely on different, but non-nested parallel trends assumptions. Second, our estimation method can be cast as a regression specification and thus may be more familiar to applied researchers. However, a third difference is that the procedure of Callaway and Sant’Anna (2020a) allows for conditioning on time-varying covariates. de Chaisemartin and D’Haultfœuille (2020) and Goodman-Bacon (2018) respectively propose alternative estimators and diagnostic tools for estimation of causal effects in staggered settings, but do not consider the estimation of the dynamic path of treatment effects as we do.

## 2 Event studies design

In this section we first formalize the “event studies design”. As discussed in Section 2.3, based on how this term is deployed in the empirical literature, an *event study* design is a *staggered adoption* design where units are treated at different times, and there may or may not be never treated units. It also nests a *difference-in-differences* design, where units are either first treated at time  $t_0$  or never treated.

Specifically, we consider a setting with a random sample of  $N$  units observed over  $T + 1$  time periods, where  $T$  is fixed. For each  $i \in \{0, \dots, N\}$  and  $t \in \{0, \dots, T\}$ , we observe the outcome  $Y_{i,t}$  and treatment status  $D_{i,t} \in \{0, 1\}$ :  $D_{i,t} = 1$  if  $i$  is treated in period  $t$  and  $D_{i,t} = 0$  if  $i$  is not treated in period  $t$ . Throughout we assume that the observations  $\{Y_{i,t}, D_{i,t}\}_{t=0}^T$  are independent and identically distributed (i.i.d.).

In the general case of event studies we focus on an *absorbing* treatment such that the treatment status over time is a non-decreasing sequence of zeros and then ones, i.e.  $D_{i,s} \leq D_{i,t}$  for  $s < t$ . We can thus uniquely characterize a treatment path by the time period of the initial treatment, denoted with  $E_i = \min \{t : D_{i,t} = 1\}$ . If unit  $i$  is never treated i.e.  $D_{i,t} = 0$  for all  $t$ , we set  $E_i = \infty$ . Based on when they first receive the treatment, we can also uniquely categorize units into disjoint cohorts  $e$  for  $e \in \{0, \dots, T, \infty\}$ , where units in cohort  $e$  are first treated at the same time  $\{i : E_i = e\}$ .

We define  $Y_{i,t}^e$  to be the potential outcome in period  $t$  when unit  $i$  is first treated in time period  $e$ . We define  $Y_{i,t}^\infty$  to be the potential outcome if unit  $i$  never receives the treatment, which we call the “baseline outcome”. Since the timing of the initial treatment uniquely characterizes one’s treatment path, we can

active literature studying inference under violations of the parallel trends assumption e.g. Rambachan and Roth (2020).

Our parallel trends assumption coincides with that of de Chaisemartin and D’Haultfœuille (2020). One could substitute this assumption with a different identifying assumption that baseline outcomes are mean independent of  $E_i$  i.e. at each  $t$ ,  $E[Y_{i,t}^\infty | E_i = e]$  is the same for all  $e \in \text{supp}(E_i)$  and in particular is equal to  $E[Y_{i,t}^\infty]$ . This stronger assumption is plausible when the timing of treatment is indeed randomized, which is the assumption used by Athey and Imbens (2018). By taking the “fully dynamic” specification as their DGP, Borusyak and Jaravel (2017) implicitly assume this version of a parallel trends assumption. Callaway and Sant’Anna (2020a) propose a weaker version that is conditional on covariates. Finally, for a particular estimand, Goodman-Bacon (2018) identifies a weaker version that only requires a weighted average of  $E[Y_{i,t}^\infty - Y_{i,s}^\infty | E_i = e]$  (averaged across cohorts) to be zero.

**Assumption 2.** (No anticipatory behavior prior to treatment.) *There is no treatment effect in pre-treatment periods i.e.  $E[Y_{i,e+\ell}^e - Y_{i,e+\ell}^\infty | E_i = e] = 0$  for all  $e \in \text{supp}(E_i)$  and all  $\ell < 0$ .*

Assumption 2 requires potential outcomes in any  $\ell$  periods before treatment to be equal to the baseline outcome on average as in Malani and Reif (2015) and Botosaru and Gutierrez (2018). This is most plausible if the full treatment paths are not known to units. If they have private knowledge of the future treatment path they may change their behavior in anticipation and thus the potential outcome prior to treatment may not represent baseline outcomes. For example, Hendren (2017) shows that knowledge of future job loss leads to decreases in consumption. If the periods with anticipation behavior are known, then we may consider an alternative version of Assumption 2, which holds for pre-periods in a subset of pre-treatment periods. Depending on the application, it may still be plausible to assume no anticipation until  $K$  periods before the treatment.

The no anticipation assumption proposed by Athey and Imbens (2018) is a deterministic condition which stipulates that  $Y_{i,e+\ell}^e = Y_{i,e+\ell}^\infty$  for all units  $i$  and  $e$  and  $\ell < 0$ . By taking the “fully dynamic” specification as their DGP, Borusyak and Jaravel (2017) allow anticipation by including pre-trends indicators in the DGP. Callaway and Sant’Anna (2020a) and Goodman-Bacon (2018) implicitly assume no anticipation by using observed outcomes in time periods before the initial treatment as the untreated potential outcomes.

**Assumption 3.** (Treatment effect homogeneity.) *For each relative period  $\ell$ ,  $CATT_{e,\ell}$  does not depend on cohort  $e$  and is equal to  $ATT_\ell$ .*

relative periods  $\ell' \notin g$ , with the same weights stated in Proposition 1:

$$\mu_g = \sum_{\ell \in g} \sum_e \omega_{e,\ell}^g CATT_{e,\ell} + \sum_{g' \neq g, g' \in \mathcal{G}} \sum_{\ell' \in g'} \sum_e \omega_{e,\ell'}^g CATT_{e,\ell'} + \sum_{\ell' \in g^{excl}} \sum_e \omega_{e,\ell'}^g CATT_{e,\ell'}. \quad (13)$$

Under Assumption 1, the terms in Proposition 1 reduce to a linear combination of the causally interpretable building blocks  $CATT_{e,\ell}$  as follows:

$$E[Y_{i,e+\ell} - Y_{i,0}^\infty | E_i] - E[Y_{i,e+\ell}^\infty - Y_{i,0}^\infty] = CATT_{e,\ell} + \underbrace{E[Y_{i,t}^\infty - Y_{i,0}^\infty | E_i] - E[Y_{i,t}^\infty - Y_{i,0}^\infty]}_{=0} \quad (14)$$

for  $t = e + \ell$ . However, two issues for interpretability remain. First, the coefficient  $\mu_g$  can be written as an average of not only  $CATT_{e,\ell}$  from own periods  $\ell \in g$ , but also  $CATT_{e,\ell'}$  from other periods. Second, the weights are still non-linear functions of the distribution of cohorts, same as those in (12), and they are not restricted to lie in  $[0, 1]$ .

The properties of these weights as described following Proposition 1 imply that contamination from other periods wanes once we impose restrictions on treatment effects. In the next two subsections we illustrate how that can happen.

### 3.4 Interpreting the coefficients under parallel trends and no anticipation assumptions

**Proposition 3.** *If Assumption 1 (parallel trends) holds and Assumption 2 (no anticipatory behavior in all periods before the initial treatment) holds, the population regression coefficient  $\mu_g$  is a linear combination of post-treatment  $CATT_{e,\ell'}$  for all  $\ell' \geq 0$ , with the same weights stated in Proposition 1:*

$$\mu_g = \sum_{\ell' \in g, \ell' > 0} \sum_e \omega_{e,\ell'}^g CATT_{e,\ell'} + \sum_{g' \neq g, g' \in \mathcal{G}} \sum_{\ell' \in g', \ell' > 0} \sum_e \omega_{e,\ell'}^g CATT_{e,\ell'} + \sum_{\ell' \in g^{excl}, \ell' > 0} \sum_e \omega_{e,\ell'}^g CATT_{e,\ell'}. \quad (15)$$

Once we restrict pre-treatment  $CATT_{e,\ell \leq 0}$  to be zero under the no anticipatory behavior assumption, the expression for  $\mu_g$  simplifies as terms involving  $CATT_{e,\ell \leq 0}$  drop out. However, the second term in the expression for  $\mu_g$  remains unless we further impose treatment effect homogeneity for its summands to cancel out each other. Thus,  $\mu_g$  may be non-zero for pre-treatment periods even if parallel trends holds.

This result immediately implies a shortcoming of using pre-treatment coefficients (i.e.  $\mu_g$  where  $g$  contains only leads to the treatment  $\ell < 0$ ) to test for pretrends. Under the no anticipatory behavior assumption,

with weight estimates from step 2. More formally, the IW estimator is

$$\widehat{v}_g = \frac{1}{|g|} \sum_{\ell \in g} \sum_e \widehat{\delta}_{e,\ell} \widehat{Pr}\{E_i = e \mid E_i \in [-\ell, T - \ell]\} \quad (27)$$

where  $\widehat{\delta}_{e,\ell}$  is returned from step 1 and  $\widehat{Pr}\{E_i = e \mid E_i \in [-\ell, T - \ell]\}$  is the estimated weight returned from step 2. We normalize the weights further by the size of  $g$ . If  $g$  is a singleton, then its size is  $|g| = 1$ .

**Validity of the IW estimator.** Under the parallel trends and no anticipation assumptions the coefficient estimator  $\widehat{\delta}_{e,\ell}$  from regression (26) is a consistent estimator for  $CATT_{e,\ell}$ . The sample shares of each cohort are also consistent estimators for the population shares. Thus, the IW estimator is consistent for a weighted average of  $CATT_{e,\ell}$  with weights equal to the share of each cohort in the relevant period(s).

With a few standard assumptions (which we present as Assumption 4 in Online Appendix B) on regression (26), we can show that each IW estimator is asymptotically normal and derive its asymptotic variance. The large sample approximation allows us to estimate the variance of IW estimators directly without relying on bootstrapping as in Callaway and Sant'Anna (2020a). However, we only construct pointwise confidence interval valid for a given IW estimator  $\widehat{v}_g$ . The bootstrap-based inference by Callaway and Sant'Anna (2020a) constructs simultaneous confidence intervals that are valid for the entire path of  $\widehat{v}_g$ .

## 4.2 Difference-in-differences estimator for $CATT_{e,\ell}$

**Definition 2.** Assume cohort  $e$  is non-empty i.e.  $\sum_{i=1}^N \mathbf{1}\{E_i = e\} > 0$ . Assume there exists some pre-period  $s < e$  and some set of control cohorts  $C \subseteq \{c : e + \ell < c \leq T\}$  that are non-empty i.e.  $\sum_{i=1}^N \mathbf{1}\{E_i \in C\} > 0$ . Using the notion  $\mathbb{E}_N$  to abbreviate the symbol  $\frac{1}{N} \sum_{i=1}^N$ , the DID estimator with pre-period  $s$  and control cohorts  $C$  estimates  $CATT_{e,\ell}$  as

$$\widehat{\delta}_{e,\ell} = \frac{\mathbb{E}_N[(Y_{i,e+\ell} - Y_{i,s}) \cdot \mathbf{1}\{E_i = e\}]}{\mathbb{E}_N[\mathbf{1}\{E_i = e\}]} - \frac{\mathbb{E}_N[(Y_{i,e+\ell} - Y_{i,s}) \cdot \mathbf{1}\{E_i \in C\}]}{\mathbb{E}_N[\mathbf{1}\{E_i \in C\}]} \quad (28)$$

The assumption of non-empty cohort  $e$ , existence of pre-period and non-empty control cohorts makes the DID estimator well-defined. For example, DID estimators for cohort 0 are not well-defined because a pre-period does not exist for this cohort, which is why we exclude them from estimating regression (26) in Step 1.

Using the above definition, in regression (26) from Step 1 of our proposed method, the coefficient

estimator  $\widehat{\delta}_{e,\ell}$  is a DID estimator for  $CATT_{e,\ell}$  with pre-period  $s = e - 1$  (because we exclude relative period  $\ell = -1$ ) and some choice of control cohorts  $C$ . If there is a never-treated cohort i.e.  $\infty \in \text{supp}\{E_i\}$ , then we set the control cohort to be never-treated units  $C = \{\infty\}$ . If there are no never-treated units, i.e.  $\infty \notin \text{supp}\{E_i\}$ , then we set the control cohort  $C = \{\max\{E_i\}\}$ , i.e. the latest-treated cohort. Among all possible pre-periods, for regression (26) from step 1 of our proposed method we choose  $s = e - 1$  and  $C = \{\infty\}$  with never-treated units (or  $\{\max\{E_i\}\}$  without never-treated units) because the resulting specification is a natural extension of the common specifications of two-way fixed effects regression.

Note that without never-treated units we need to drop time periods  $t \geq \max\{E_i\}$  from estimating regression (26) because every unit will be treated in these periods. The DID estimators for  $CATT_{e,\ell}$  for  $e + \ell \geq \max\{E_i\}$  are thus not well-defined as the control cohort is empty  $C = \emptyset$ . For example, when there are just two cohorts with  $E_i \in \{1, T\}$ , one treated at  $t = 1$  and the other treated in the last period, we then need to omit interaction terms involving the latest-treated cohorts as well as dropping observations from the last period  $t = T$  from estimation.

Under some assumptions, this DID estimator  $\widehat{\delta}_{e,\ell}$  is an unbiased and consistent estimator for  $CATT_{e,\ell}$ , a fact that we build on in deriving the probability limit of the IW estimator. We state this in the following proposition.

**Proposition 5.** *If Assumptions 1 and 2 hold, then the DID estimator using any pre-period  $s < e$  and non-empty control cohorts  $C$  is an unbiased and consistent estimator for  $CATT_{e,\ell}$ .*

It is possible to relax the parallel trends assumption to allow the timing of treatment to depend on covariates. One can estimate  $CATT_{e,\ell}$  consistently based on the inverse propensity score reweighted estimator proposed by Abadie (2005) and Callaway and Sant’Anna (2020a), the outcome regression approach by Heckman et al. (1997), and the doubly robust estimator recently proposed by Sant’Anna and Zhao (2020). The resulting estimates  $\widehat{\delta}_{e,\ell}$  can then be plugged into step 3 to form our IW estimator. In particular, without covariates and for the case with never treated units, our approach coincides with Callaway and Sant’Anna (2020a). Therefore one can use the did R package developed by Callaway and Sant’Anna (2020b) to form the IW estimator.

The choice of the pre-period  $s$  and the control cohorts  $C$  depends on the trade-off between relaxing Assumption 1 or Assumption 2. If Assumption 2 is likely to hold, we can choose any pre-period  $s < e$  and include not-yet-treated units in control cohorts  $C = \{c : c > e + \ell\}$  as in Callaway and Sant’Anna (2020a). This